

Chapter 19

Diversity and Distribution of Reef Fishes in Gulf of Mannar Islands, India

J. S. Yogesh Kumar, S. Geetha and R. Sornaraj

Abstract Reef fishes in the coral reef habitat of Thoothukudi and Vembar group of islands in Gulf of Mannar, India are studied during the period from July 2007 to 2008. Overall 117 species under 79 genera, 42 families were recorded by the visual census method. Among them 29 species were noticed in all study stations. The abundance of in the study area was estimated for fishes Lethrinidae (11 and 10.6 %), Chaetodontidae (9.9 and 11.1 %), and Pomacentridae (8 and 9.2 %) family was dominated in all the study stations followed by Labridae (8.4 and 6.9 %), Serranidae (8.7 and 6.4 %), Apogonidae (6.6 and 5.8 %) at Thoothukudi and Vembar group of islands, respectively. The highest species diversity, richness and evenness were recorded at Upputhanni Islands and lowest values for these variables were reported at Kaswori Island during the study period. The study revealed high coral reef fish density in the area indicating the need for adopting conservation measures.

Keywords Abundance • Distribution • Diversity • Gulf of Mannar • Reef fish

19.1 Introduction

Coral reef habitat is a complex ecosystem providing shelter to many animals including nudibranchs, molluscs, echinoderms, crustaceans, sponges, fishes, and other marine vertebrates. Discrete population, small breeding pools, sedentary and

J. S. Yogesh Kumar (✉)

Zoological Survey of India, Coral Transplantation and Restoration Project
(Gulf of Kutch), “Van Sakul” Nr. Nagnath Gate, Jamnagar, Gujarat, India
e-mail: coralyogesh@yahoo.co.in

S. Geetha

Wetland Research and Development, Thoothukudi, Tamil Nadu, India

R. Sornaraj

Kamaraj College, Thoothukudi, Tamil Nadu, India

non-migratory nature and association with specific habitats are striking characters of coral reef fishes. A variety of habitats found in coral reef areas may be rich or poor in species in each habitat or between differing habitats, supporting markedly different fish communities of which some may be cosmopolitan. Depending on a particular food item, behavioral interactions and depths across reefs are other factors known to limit reef fish distribution (Turner et al. 2001). Reef fishes are the most diverse elements in the reef fauna and because of their wider ecological significance, some families of reef fish present valuable groups for monitoring the health of reefs and for investigating factors underlying the high species diversity characteristics of reef ecosystems. Some fishes such as species of butterfly fish have been proposed as useful indicator species of reef development as well as health (Reese 1981, Ohman et al. 1998). Resources in the reefs are fast dwindling and hence the study of diversity in the coral reef ecosystem is of great significance in order to assess the changes over the period of time. The strength of association between organisms and their habitat can provide an indication of the level of habitat change (Jones and Andrew 1993) and an array of studies have documented positive relationships between fish abundance as well as diversity and coral cover (Bell and Galzin 1984; Findley and Findley 1985; Hart et al. 1996).

Gulf of Mannar has been one of the leading producers of marine fish in the country. Of late, however, the effect of increased fishing pressure on the resources with increasing demand and modernization of fishing practices have resulted in a decrease in the catches of several highly valued resources (Asha et al. 2009). In Gulf of Mannar, a total of 903 reef fish reported, of which 125 species are commercially important finfishes while 115 marine ornamental fishes (Venkataramani et al. 2005). The biodiversity and biology of these families have been studied in detail in the Gulf of Mannar Province (Venkataramani and Jawahar 2004).

The present scenarios of changing environmental conditions are threat to the patchy reefs around the Thoothukudi and Vembar group of islands. Knowledge of the current status of fish communities coupled with long-term monitoring is essential for formulating future management strategies. Hence, this study was initiated to assess the fish diversity of the area.

19.2 Methods

19.2.1 Study Area Profile

The Gulf of Mannar Biosphere Reserve (GoMBR) encompasses 21 islands between Rameshwaram and Thoothukudi (08°47'N 78°12'E and 09°15'N 79°14'E). The Gulf of Mannar was declared as marine national park in 1986 for the purpose of protecting marine Wild life and its environment (Upreti and Shanmugaraj 1997). These are uninhabited islands, ranging in size from 0.25 to 130 ha, along the coast of 170 km, with the closest being 500 meters from shore

and farthest, over 4 km. The island and their shallow waters form the core of the reserve. This core area is in turn surrounded by a 10 km wide buffer zone. The islands in the Gulf of Mannar are classified into four major groups.

1. *Mandapam group* (seven islands): Musal, Manoli, Manoliputti, Poomarichan Pullivasal, Krusadai, and Shingle.
2. *Keezhakkarai group* (seven islands): Yaanaipar, Vallimunai, Poovarasaputti, Appa, Thalaiyari, Vaalai, and Mulli.
3. *Vembar group* (three islands): Upputhanni, Pulivinichalli, and Nallathanni.
4. *Thoothukudi group* (four islands): Van, Koswari, Kariyachalli, and Velanguchalli (Submerged).

However, this study was conducted in Thoothukudi and Vembar group of islands.

19.2.2 Visual Census Method (Belt Transect)

Reef fish abundance estimates were done using underwater visual census technique using belt transect by SCUBA diving (Brock 1954). For this study, the selected transect dimensions were 30 m length by 5 m from each side of the transect line (30 × 10 m). A total of nine transects were done in all the sites. The fish observer swam slowly along the transect of each site to identify and count the fishes located within an area of 300 m² by visual estimates following English et al. (1997). Underwater photo of fishes were identified to species level following Randall et al. (1996) and Lieske and Myers (1996). During the survey, we analyzed the difference of diversity between the study sites, Shannon–Weiner diversity index (H') and Pielou's Evenness index (J') for evenness were calculated (Ludwig and Reynolds 1988).

19.3 Results

The species composition for each station is presented in Table 19.1. The number of species observed on both sides of belt transect were 41 species from Van Island, 38 species from Koswari Island, 73 species from Kariyachalli Island, 84 have been recorded from Upputhanni Island, 52 species from Pulivinichalli Island, and 57 species from Nallathanni Island. Overall 117 species under 79 genera, 42 families were reported [Figs. 19.1, 19.4(1–6), 19.5(7–12), 19.6(13–18)]. From the observations, it was found that 29 species of reef fish were noticed in all study stations (*Cheilodipterus quinquelineatus*, *Rhabdamia gracilis*, *Caranx melampygus*, *Chaetodon colare*, *Chaetodon deccusatus*, *Chaetodon octofasciatus*, *Chaetodon andamanensis*, *Heniochus acuminatus*, *Cirrhitichthys bleekeri*, *Plectorhinchus*

Table 19.1 Reef fishes recorded from Thoothukudi and Vembar group of islands, Gulf of Mannar

Sl. no.	Family	Species	1	2	3	4	5	6
1.	Acanthuridae	<i>Acanthurus grammoptilus</i> Richardson, 1843	+	+		+		
2.		<i>Acanthurus leucosternon</i> Bennett 1832				+	+	+
3.	Apogonidae	<i>Apogon angustatus</i> (Smith and Radcliffe, 1911)				+		+
4.		<i>Apogon fragilis</i> Smith, 1961	+	+	+		+	+
5.		<i>Apogon griffini</i> (Seale, 1910)				+		
6.		<i>Archamia fucata</i> (Cantor, 1850)		+			+	
7.		<i>Cheilodipterus quinquelineatus</i> Cuvier, 1828	+	+	+	+	+	+
8.		<i>Rhabdamia gracilis</i> (Bleeker, 1856)	+	+	+	+	+	+
9.	Atherinidae	<i>Atherinomorus lacunosus</i> (Bloch and Schneider, 1801)				+		
10.	Balistidae	<i>Pseudobalistes flavimarginatus</i> (Ruppell, 1829)					+	
11.		<i>Sufflamen chrysopterus</i> (Bloch and Schneider, 1801)	+					
12.		<i>Sufflamen frenatus</i> (Latreille, 1804)					+	
13.	Blenniidae	<i>Plagiotremus rhinorhynchus</i> (Bleeker, 1852)				+		
14.		<i>Mimoblennius atrocinctus</i> (Regan, 1909)					+	
15.	Bothidae	<i>Bothus pantherinus</i> (Ruppell, 1830)		+	+			+
16.	Caesionidae	<i>Pterocaesio tessellata</i> Carpenter, 1987				+	+	+
17.	Carangidae	<i>Caranx ignobilis</i> (Forsskål, 1775)					+	
18.		<i>Caranx melampygus</i> Cuvier, 1833	+	+	+	+	+	+
19.		<i>Gnathanodon speciosa</i> Post, 1973				+	+	
20.	Chaetodontidae	<i>Chaetodon auriga</i> Forsskal, 1775				+	+	+
21.		<i>Chaetodon colare</i> (Bloch, 1787)	+	+	+	+	+	+
22.		<i>Chaetodon deccusatus</i> Cuvier, 1829	+	+	+	+	+	+
23.		<i>Chaetodon lineolatus</i> Cuvier, 1831	+		+	+	+	+
24.		<i>Chaetodon octofasciatus</i> (Bloch, 1787)	+	+	+	+	+	+
25.		<i>Chaetodon andamanensis</i> Kuiter and Debelius, 1999	+	+	+	+	+	+
26.		<i>Chaetodon trifasciatus</i> Park, 1797				+	+	
27.		<i>Chaetodon xanthocephalus</i> Bennett, 1833					+	
28.		<i>Heniochus acuminatus</i> (Linnaeus, 1758)	+	+	+	+	+	+
29.		<i>Heniochus singularius</i> Smith and Radcliffe, 1911					+	
30.	Cirrhitidae	<i>Cirrhitichthys bleekeri</i> Day, 1874	+	+	+	+	+	+
31.	Congridae	<i>Gorgasia maculata</i> Klauswitz and Eibl-Eibesfeldt 1959				+		
32.	Diodontidae	<i>Diodon hystrix</i> Linnaeus, 1758				+	+	
33.	Echeneidae	<i>Echeneis naucrates</i> Linnaeus, 1758					+	
34.	Eleotridae	<i>Brycaninops yongei</i> (Davis and Cohen, 1968)				+		

(continued)

Table 19.1 (continued)

Sl. no.	Family	Species	1	2	3	4	5	6
35.	Ephippidae	<i>Platax pinnatus</i> (Linnaeus, 1758)				+		
36.		<i>Zabidius novemaculeatus</i> (McCulloch, 1916)			+			
37.	Gobiidae	<i>Amblygobius semicinctus</i> (Bennett, 1833)			+			
38.		<i>Cryptocentrus cinctus</i> (Herre, 1936)			+	+	+	
39.		<i>Valenciennesa sexguttata</i> (Valenciennes, 1837)			+	+		
40.	Haemulidae	<i>Diagramma pictum</i> (Thunberg, 1792)		+	+	+		
41.		<i>Plectorhinchus flavomaculatus</i> (Cuvier, 1830)		+	+		+	
42.		<i>Plectorhinchus vittatus</i> (Linnaeus, 1758)	+	+	+	+	+	+
43.		<i>Plectorhynchus unicolor</i> (Macleay 1883)						+
44.	Holocentridae	<i>Sargocentron rubrum</i> (Forsskal, 1775)	+	+	+	+	+	+
45.	Labridae	<i>Bodianus axillaris</i> (Bennett, 1831)			+			
46.		<i>Cheilinus chlorourus</i> (Bloch, 1791)			+			
47.		<i>Coris formosa</i> (Bennett, 1830)			+			+
48.		<i>Halichoeres hartzfeldii</i> (Bleeker, 1852)	+		+		+	
49.		<i>Hemigymnus melapterus</i> (Bloch, 1791)						+
50.		<i>Labroides dimidiatus</i> (Valenciennes, 1839)	+	+	+	+	+	+
51.		<i>Leptojulius cyanopleura</i> (Bleeker, 1853)				+		
52.		<i>Thalassoma janseni</i> (Bleeker, 1856)				+		+
53.		<i>Thalassoma lunare</i> (Linnaeus, 1758)				+		
54.		<i>Xyrichtys pentadactylus</i> (Linnaeus, 1758)				+		
55.	Labridae	<i>Cheilinus undulatus</i> Ruppell, 1835			+			
56.	Lethrinidae	<i>Lethrinus harak</i> (Forsskal, 1775)		+		+	+	+
57.		<i>Scolopsis torquatus</i> (Cuvier 1830)				+		
58.		<i>Caesio caeruleaurea</i> Lacepede, 1801	+	+	+	+	+	+
59.		<i>Caesio cuning</i> (Bloch, 1791)				+		
60.		<i>Lutjanus argentimaculatus</i> (Forsskal, 1775)						+
61.		<i>Lutjanus fulviflamma</i> (Forsskal, 1775)			+	+		
62.		<i>Lutjanus fulvus</i> (Bloch and Schneider, 1801)	+	+	+	+	+	+
63.		<i>Lutjanus lutjanus</i> Bloch, 1790	+	+	+	+	+	+
64.		<i>Lutjanus quinquelineatus</i> (Bloch, 1790)	+	+	+	+	+	+
65.	Microdesmidae	<i>Ptereleotris microlepis</i> (Bleeker, 1856)			+			
66.	Monodactylidae	<i>Monodactylus argenteus</i> (Linnaeus, 1758)			+		+	+
67.	Mullidae	<i>Parupeneus cyclostomus</i> (Lacepede, 1801)						+
68.		<i>Parupeneus indicus</i> (Shaw, 1803)			+			
69.		<i>Upeneus tragula</i> Richardson, 1846	+		+	+	+	+
70.	Muraenidae	<i>Gymnothorax permistus</i> (Smith, 1962)				+		
71.		<i>Gymnothorax pseudothyrsoides</i> (Bleeker, 1852)			+			+
72.		<i>Siderea thyrsoides</i> (Richardson, 1845)				+		

(continued)

Table 19.1 (continued)

Sl. no.	Family	Species	1	2	3	4	5	6
73.	Ostraciidae	<i>Ostracion cubicus</i> Linnaeus, 1758			+	+	+	+
74.		<i>Rhynchostracion rhinorhynchus</i> (Bleeker, 1852)				+		
75.	Pegasidae	<i>Pegasus volitans</i> Linnaeus, 1758				+		
76.		<i>Eurypegasis draconis</i> (Linnaeus, 1766)				+		
77.	Pempheridae	<i>Pempheris adusta</i> Bleeker, 1877			+	+		+
78.		<i>Pempheris oualensis</i> Cuvier, 1831	+				+	
79.	Pinguipedidae	<i>Parapercis lineopunctata</i> Randall, 2003	+	+	+	+	+	+
80.	Plotosidae	<i>Paraplotosus albilabrus</i> (Valenciennes, 1840)				+		
81.	Pomacanthidae	<i>Apolemichthys xanthurus</i> (Bennett, 1833)				+	+	+
82.		<i>Centropyge flavipectoralis</i> Randall and Klauswitz, 1977	+	+	+	+	+	+
83.		<i>Centropyge multispinus</i> (Playfair and Gunther 1867)	+			+		+
84.		<i>Pomacanthus annularis</i> (Bloch 1787)				+		
85.		<i>Pomacanthus semicirculatus</i> (Cuvier 1831)	+	+	+	+	+	+
86.		<i>Pomacanthus sexstriatus</i> (Cuvier, 1831)				+		
87.	Pomacentridae	<i>Abudefduf bengalensis</i> (Bloch, 1787)				+		+
88.		<i>Abudefduf saxatilis</i> (Linnaeus, 1758)				+		
89.		<i>Amphiprion sebae</i> Bleeker, 1853	+	+	+	+	+	+
90.		<i>Dascyllus trimaculatus</i> (Ruppell, 1828)			+	+	+	+
91.		<i>Neopomacentrus azysron</i> (Bleeker, 1877)				+		
92.		<i>Neopomacentrus cyanomos</i> (Bleeker, 1856)		+	+		+	
93.		<i>Pomacentrus brachialis</i> Cuvier, 1830	+	+	+		+	+
94.		<i>Pomacentrus proteus</i> (Allen 1991)	+	+	+	+	+	+
95.		<i>Pomacentrus similis</i> Allen 1991			+	+		+
96.	Scaridae	<i>Chlorurus oedema</i> (Snyder, 1909)				+		
97.		<i>Scarus caudofasciatus</i> (Gunther, 1862)				+		
98.	Scorpaenidae	<i>Pterois miles</i> (Bennett, 1828)			+	+		+
99.		<i>Pterois volitans</i> (Linnaeus, 1758)	+		+		+	
100.		<i>Scorpaenopsis venosa</i> (Cuvier, 1829)			+			
101.	Serranidae	<i>Cephalopholis formosa</i> (Shaw, 1804)	+	+	+	+	+	+
102.		<i>Epinephelus caeruleopunctatus</i> (Bloch, 1790)	+	+	+	+	+	+
103.		<i>Epinephelus coioides</i> (Hamilton, 1822)	+		+	+	+	+
104.		<i>Epinephelus malabaricus</i> (Bloch and Schneider, 1801)	+		+	+		
105.		<i>Epinephelus tauvina</i> (Forsskal, 1775)	+	+	+	+	+	+
106.	Siganidae	<i>Siganus javus</i> (Linnaeus, 1766)		+	+			
107.	Sphyraenidae	<i>Sphyraena jello</i> Cuvier, 1829				+		
108.	Syngnathidae	<i>Corythoichthys haematopterus</i> (Bleeker, 1851)	+	+	+	+	+	+
109.		<i>Hippocampus kuda</i> Bleeker, 1852				+		+
110.		<i>Trachyrhamphus bicoarctatus</i> (Bleeker, 1857)	+	+	+	+	+	+

(continued)

Table 19.1 (continued)

Sl. no.	Family	Species	1	2	3	4	5	6
111.	Synodontidae	<i>Synodus binotatus</i> Schultz, 1953	+	+	+	+	+	+
112.	Tetraodontidae	<i>Arothron immaculatus</i> (Bloch and Schneider, 1801)	+			+	+	+
113.		<i>Canthigaster papua</i> (Bleeker, 1858)				+	+	
114.		<i>Canthigaster solandri</i> (Richardson, 1845)			+	+		
115.	Torpedinidae	<i>Dasyatis kuhlii</i> (Muller and Henle, 1841)	+	+	+	+	+	+
116.	Tripterygiidae	<i>Enneapterygius elegans</i> (Peters, 1877)				+		
117.	Zanclidae	<i>Zanclus cornutus</i> (Linnaeus, 1758)				+	+	

(1 Van, 2 Koswari, 3 Kariyachalli, 4 Upputhanni, 5 Pulivinichalli and 6 Nallathanni)

vittatus, *Sargocentron rubrum*, *Labroides dimidiatus*, *Scolopsis vosmeri*, *Caesio caeruleaurea*, *Lutjanus fulvus*, *Lutjanus lutjanus*, *Lutjanus quinquelineatus*, *Parapercis cylindrical*, *Centropyge flavipectoralis*, *Pomacanthus semicirculatus*, *Amphiprion sebae*, *Pomacentrus proteus*, *Cephalopholis Formosa*, *Epinephelus caeruleopunctatus*, *Epinephelus tauvina*, *Corythoichthys haematopterus*, *Trachyrhamphus bicoarctatus*, *Synodus binotatus*, and *Dasyatis kuhlii*).

Reef fish's abundance was compared with family wise between these two groups of islands (Fig. 19.5). The fishes of Lethrinidae (11 ± 2.9 , 10.6 ± 0.7 %), Chaetodontidae (9.9 ± 0.9 , 11.1 ± 0.7 %), and Pomacentridae (8 ± 1.5 , 9.2 ± 0.6 %) family was dominated in all the study stations followed by Labridae (8.4 ± 1.4 , 6.9 ± 1.3 %), Serranidae (8.7 ± 2 , 6.4 ± 0.6 %), Apogonidae (6.6 ± 2.6 , 5.8 ± 3.2 %) represented at Thoothukudi and Vembar group of islands, respectively.

The diversity indices of fishes at Thoothukudi and Vembar groups, the variance of species diversity, evenness and richness are shown in Fig. 19.6. The highest species diversity, richness and evenness were recorded at Upputhanni Islands (5.29, 0.83, and 0.95) and lowest values for these variables were reported at Kaswori Island (4.49, 0.84, and 0.93) during the study period.

19.4 Discussion

Fish communities are a major resource of coral reef. They play an important role in coral reef environment and are commercially important for both fisheries and more recently tourism. Venkataramani and Jawahar (2004) had recorded 113 species that belong to 24 families of highly valued marine ornamental fishes from Gulf of Mannar. In this study, 117 species that belong to 42 families were recorded from this coast. Considering the number of species recorded in the survey, largest

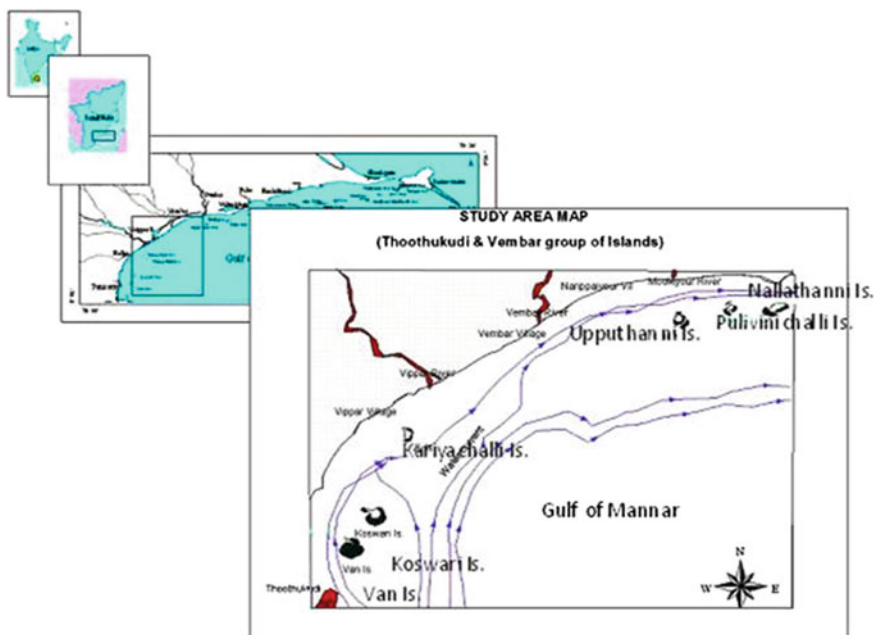


Fig. 19.1 Map showing the study areas

group that belong to Lethrinidae, Chaetodontidae, and Pomacentridae family followed by Labridae, Serranidae, Apogonidae found at Thoothukudi and Vembar group of islands. The reef fishes are diverse and abundance in the area. There exists a close relationship between the amount of reef present in the area and number of fish species occurring in that area (Hiatt and Strasburg 1960; Hobson 1974; Sale 1980; Russ 1984). As the diversity of the coral fishes is high, it could be assumed that there exists a good corals reef system in the area (Shulman 1985; Vijay Anand and Pillai 2003).

Moreover, studies in the area have also revealed the existence of patchy reefs around the island, which comprised mainly of corals of the family Poritidae, Acroporiidae, Faviidae, and Pocilloporidae. The fish species that belong to family Chaetodontidae, which is generally used as bioindicators of health of the coral and habitat disturbances, were also represented in good number on the reef (Roberts and Ormond 1987). Family Pomacentridae was the next dominant group in the area, and most of this species under this family strongly restrict themselves to specific site and have only small territory (Sale 1974, 1978; Fricke 1975). Bell et al. (1985) has suggested that rich and abundance of fishes were observed at sites rich in coral reef. Sano et al. (1984) in a series of experiments using living and dead branching corals reported that, dead colonies support fewer individuals and species of resident fishes than do live colonies.

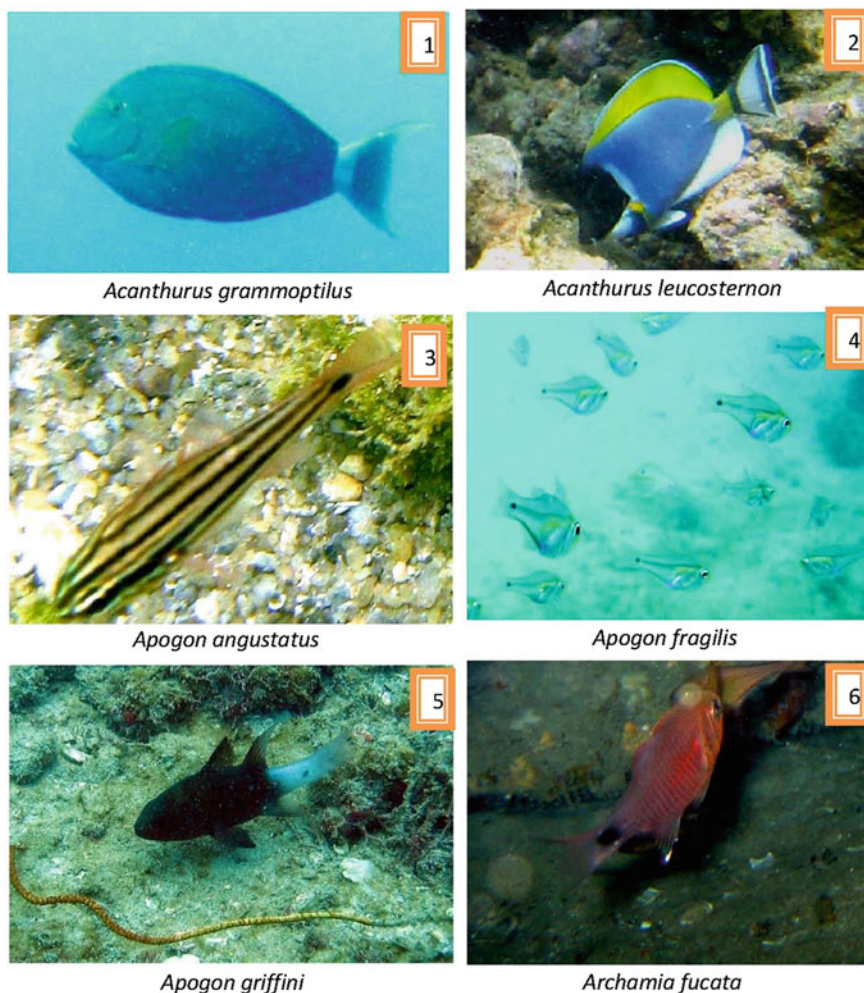


Fig. 19.2 Reef fishes from Gulf of Mannar

From this study on the reef fishes, it is evident that Gulf of Mannar coast has a good coral cover with high fish density. The role of these species in the coral reef system is limited, and it is difficult to predict the diversity loss due to degradation of the reef. However, it is clear that fishes are important to many processes on coral reef ecosystem such as control of algal populations, bioerosion and sediment production, nutrient cycling between adjacent habitats and control of populations of other important species such as echinoderms (Ormond et al. 1991; Roberts 1995). Ormond and Roberts (1999) suggested that functional complementarities of species in diverse reef system can ameliorate the impacts of species loss. There is a rapid decline in coral reef ecosystem and about 10 % of the reefs are already

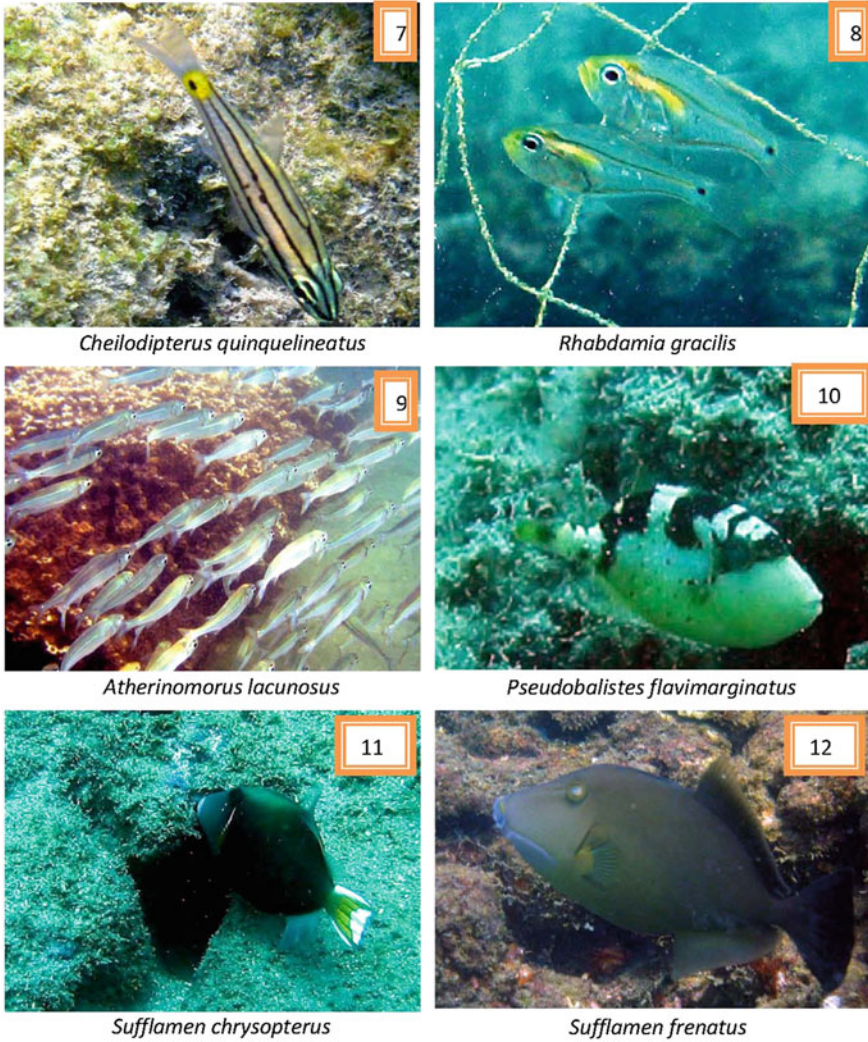


Fig. 19.3 Reef fishes from Gulf of Mannar

destroyed and 60 % are under threat globally due to various environmental and human interventions (Gray 1997). Mortality of fishes due to ecological and climatic changes have been reported previously from Mandapam group of islands Gulf of Mannar (Badrudeen et al. 1987) and recently Asha et al. 2009 reported fish mortality due to industrial waste in Thoothukudi Bay. Since uncertainties are there in the species interactions and loss of diversity in the reefs the most effective course to adopt is conservation of the ecosystem.

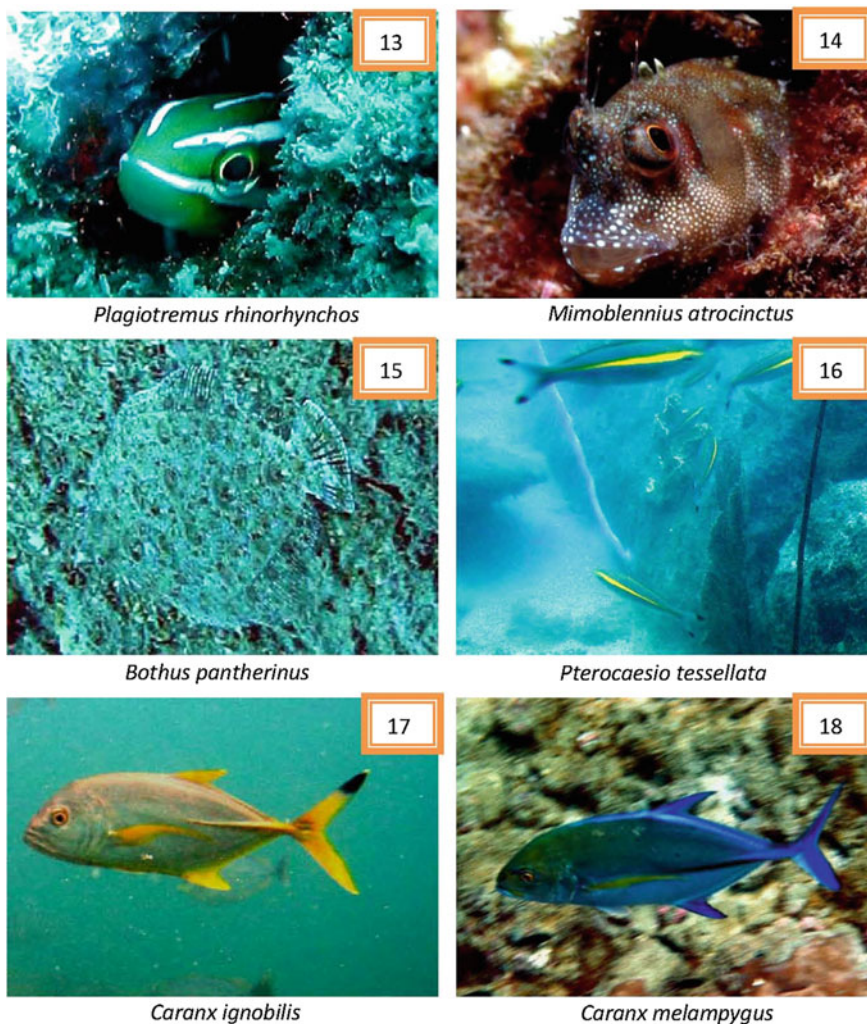


Fig. 19.4 Reef fishes from Gulf of Mannar

Reef fishes are important to many processes occurring in the reef and loss would affect the health of the coral reef ecosystem. This study revealed rich and varied diversity of reef fishes around the islands. Degradation of the reef habitat and overharvesting, industrialization is leading to heavy losses in reef diversity around the globe and the study is a pioneering effort in this line, which could form a baseline for further studies.

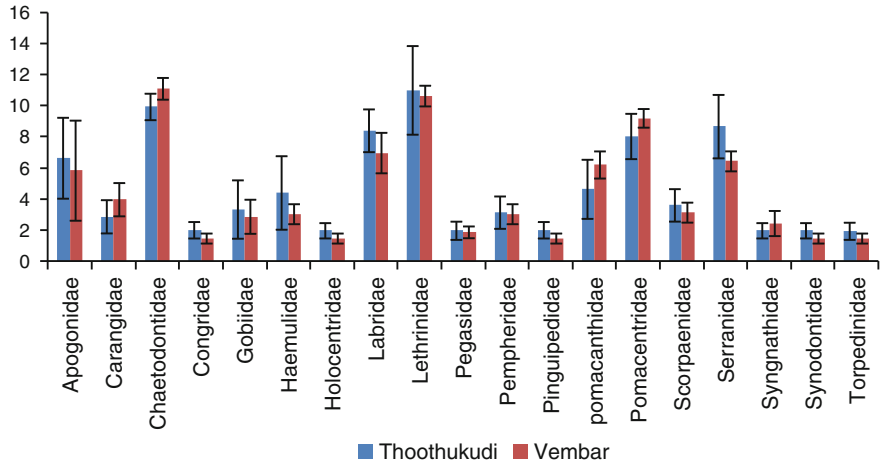


Fig. 19.5 Common reef fishes family (% with stranded deviations) differences between the Thoothukudi and Vembar group of islands at Gulf of Mannar

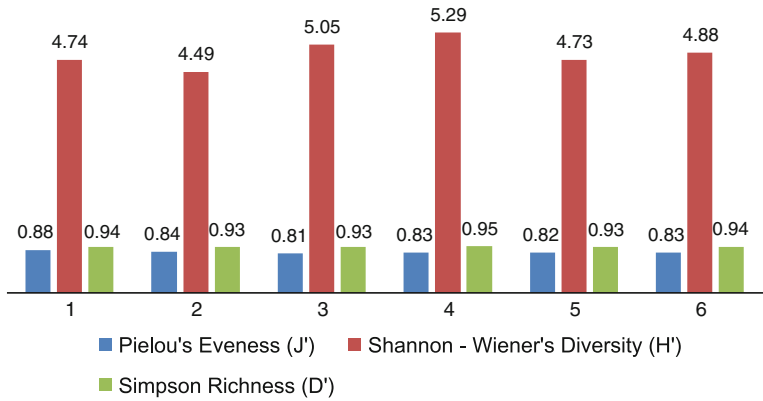


Fig. 19.6 Diversity indices for six islands at Thoothukudi and Vembar groups of Gulf of Mannar

Acknowledgments Authors wish to thank Shri. Senbagamoorthy, Department of Forest, Ramnad District for the help they rendered for permission to survey the islands. Authors also wish to express their sincere gratitude to the staff of Peoples Action for Development, NGO, Vembar especially to the Executive Director Mr. Rajendraprasad for the help in field surveys. One of the authors (J. S. Yogesh Kumar) is grateful to Dr. K. Venkataraman, Director, ZSI, Kolkata, for giving the necessary facilities to do this research study. Due gratitude is also expressed to the Principal of Kamaraj college, Thoothukudi for the help rendered.

References

- Asha PS, Joshi KK, Diwakar K (2009) Incidence of fish mortality in Thoothukudi Bay, Gulf of Mannar. *J Mar Biol Ass India* 51(2):173–177
- Badrudeen M, Fernando AB, Kasinathan C, Kaliaperumal K, Pillai SK, Pillai VK, Pandian G, Sivasdas M, Rao K (1987) An instance of mass fish mortality at Mandapam, South East Coast of India. *Mari Fish Infor Serv T E Ser* 75:1–5
- Bell JD, Galzin R (1984) Influence of live coral cover on coral-reef fish communities. *Mar Ecol Prog Ser* 15(3):265–274
- Bell JS, Craik GJS, Pollard DA, Russell BC (1985) Estimating length frequency distribution of large reef fish underwater. *Coral Reef* 4:41–44
- Brock VF (1954) A preliminary report on a method of estimating reef fish populations. *J Wildl Manage* 18:299–308
- English S, Wilkinson C, Baker V (1997) Survey manual for tropical marine resources. Australian Institute of Marine Science, Townsville 390pp
- Findley JS, Findley MT (1985) A search for pattern in butterfly fish communities. *Am Nat* 126(6):800–808
- Fricke HW (1975) The role of behaviour in marine symbiotic animals. In: Jennings DH, Lee DL (eds) *Symbiosis, symposia of the society for experimental biology*, 29. Cambridge University Press, Cambridge, pp 581–594
- Gray JS (1997) Marine biodiversity: patterns, threats and conservations needs. *Rep Stud GESAMP* 62:24pp
- Hart AM, Klumpp DW, Russ GR (1996) Response of herbivorous fishes to crown-of-thorns starfish *Acanthaster planci* outbreaks II density and biomass of selected species of herbivorous fish and fish-habitat correlations. *Mar Ecol Prog Ser* 132(1–3):21–30
- Hiatt RW, Strasburg DW (1960) Ecological relationships of the fish fauna on coral reefs of the Marshall Islands. *Ecol Monogr* 30:65–127
- Hobson ES (1974) Feeding relationships of teleostean fishes on coral reefs in Kona. *Hawaii Fish Bull* 72(4):915–1031
- Jones GP, Andrew NL (1993) Temperate reefs and the scope of seascape ecology. In: Battershill CN, Scheil DR, Jones GP, Creese RG, Mac Diarmid AB (eds) *In: Proceedings of the second international temperate reef Symposium, 7–10 Jan 1992, Auckland, New Zealand NIWA, Marine Wellington*, pp 63–76
- Lieske E, Myers R (1996) *Coral reef fishes Indo-Pacific and Caribbean*. Harper Collins Publishers, London 400p. ISBN 0002199742
- Ludwig JA, Reynolds JF (1988) *Statistical ecology*. Wiley, New York, 337pp
- Ohman MC, Rajasuriya A, Svensson S (1998) The use of butterfly fishes (Chaetodontidae) as bioindicators of habitat structure and human disturbances. *Ambio* 27:708–716
- Ormond RFG, Roberts CM (1999) Biodiversity of coral reef fishes. In: Rupert FG, Ormond J, Gage D, MV Angel (eds) *Marine biodiversity patterns and processes*, pp 216–257
- Ormond RFG, Brabury R, Bainbridge S, Fabricius K, Keesing J, De Vantier L, Medley P, Steven A (1991) Test of a model of regulation of Crown-of Thorns starfish by fish predators, In: Bradbury R (ed) *Acanthaster and the coral reef a theoretical perspective*. Springer, Berlin, pp 189–207
- Randall JE, Gerald R, Roger RA, Steene C (1996) *Fishes of the Great Barrier Reef and coral sea*. University of Hawaii Press, Honolulu 557p
- Reese ES (1981) Predation on coral by fishes of family Chaetodontidae: implications for conservation and management of coral reef ecosystems. *Bull Mar Sci* 31:594–604
- Roberts CM (1995) Effects of fishing on coral reefs—a massive uncontrolled experiment with ecosystem structure. *Conserv Biol* 9:988–995
- Roberts CM, Ormond RFG (1987) Habitat complexity and coral reef fish diversity abundance on red sea fringing reefs. *Mar Ecol Prog Ser* 41:1–8

- Russ GR (1984) Distribution and abundance of herbivorous grazing fishes in the central Great Barrier Reef. II. Patterns of zonation of mid-shelf and outer shelf reefs. *Mar Ecol Prog Ser* 20:35–44
- Sale PF (1974) Mechanism of co-existence in a guild of territorial fishes at Heron Island. In: Cameron AM, Campbell BM, and Cribb AB (eds). In: Proceedings of the second international coral reef Symposium brisbane. The Great Barrier Reef Committee, vol 1, pp 193–206
- Sale PF (1978) Coexistence of coral reef fishes—a lottery for living space. *Env Biol Fish* 3:85–102
- Sale PF (1980) The ecology of fishes on coral reefs. *Oceanogr Mar Biol* 18:367–421
- Sano M, Shimizu M, Nose Y (1984) Changes in structure of coral reef fish communities by destruction of hermatypic corals—observational and experimental views. *Pac Sci* 38:51–79
- Shulman MJ (1985) Coral reef fish assemblages: Intra-and interspecific competition for shelter sites. *Environ Biol Fishes* 13:81–92
- Turner MG, Gardner RH, O'Neil IRV (2001) Landscape ecology in theory and practice: pattern and process. Springer, Berlin 401pp
- Upreti A, Shanmugaraj T (1997) Gulf of Mannar marine biosphere reserve. 47pp
- Venkataramani VK, Jawahar V (2004) Resource assessment of ornamental reef fisheries of Gulf of Mannar, Southeast coast of India. Final report—ICAR/NATP/CGP/Project. 66p
- Venkataramani VK, Jawahar P, Vaitheeswaran T, Santhanam R (2005) Marine ornamental fishes of Gulf of Mannar. ICAR/NATP/CGP/publication. 175pp
- Vijay Anand PE, Pillai NGK (2003) Habitat distribution and species diversity of coral reef fishes in the reef slope of the Kavaratti atoll, Lakshadweep. *India J Mar Biol Ass India* 45:88–98